

About This Book

Conformal prediction offers a distribution-free way to attach a reliable uncertainty guarantee to a machine-learned predictor – but the standard guarantee holds only on average, with no promise for any specific operating condition. This book tests that limitation in a domain where discrete, queueing-driven dynamics differ structurally from where the technique was first validated: a discrete-event simulation of a hospital emergency department, calibrated on real arrival and triage-acuity data.

A gradient-boosting surrogate stands in for the simulator across a grid of staffing and arrival-rate regimes. Standard conformal prediction covers only 68.2% of the worst-affected regime against a 90% target. Mondrian conformal prediction – calibrating separately within each staffing × arrival-rate category rather than pooling – closes that gap to 90.9%, without disturbing the calibration set's exchangeability guarantee.

Five further extensions, from weighted calibration to a queueing-theoretic continuous alternative, follow – evaluated with the same honesty, including where they fall short.

THE MONDRIAN PARTITION

Arrival-rate tercile → Staffing tercile ↓



 Understaffed + high-arrival: the worst-covered category (68.2% → 90.9%)

BEYOND MARGINAL GUARANTEES

Beyond Marginal Guarantees

Mondrian Conformal Prediction for High-Variance Discrete-Event Queueing Systems

THE PIPELINE THIS BOOK BUILDS



1

EMERGENCY DEPARTMENT

Stochastic arrivals & triage acuity



2

DISCRETE-EVENT SIMULATION

Queueing dynamics & performance metrics



3

SURROGATE MODEL

Gradient-boosting regressor



4

UNCERTAINTY QUANTIFICATION

Prediction intervals with a coverage guarantee



5

MONDRIAN CONFORMAL PREDICTION

Conditional coverage within each regime